6.4.1 How LangChain agents work

Figure 6.7 illustrates the iterative workflow of LangChain agents, showing how they reason, act, and observe in a continuous loop until a final response is generated.

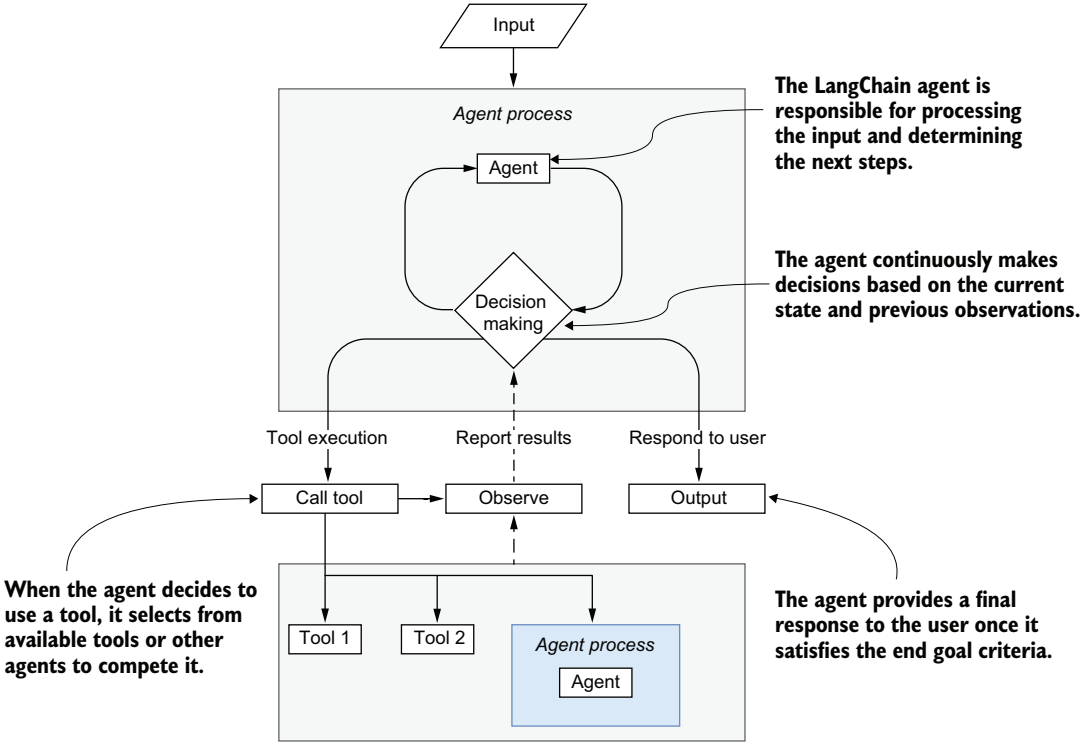


Figure 6.7 The iterative decision-making process of a LangChain agent. Starting with user input, the agent cycles through decision making, tool execution, and observation phases. The agent can utilize multiple tools (represented as tool 1, tool 2, or other agents) as needed. This process continues until the agent determines it can provide a final response to the user.

To see a better picture of this process, let's look at an example interaction for an agent that plans vacations. We start with the user prompt "Plan a seven-day vacation to Japan for a family of four with a budget of $5000." Here are the iterative steps that the agent will follow (with the tools used in italics):

1. The LangChain *agent* processes this request and begins its decision-making process.
2. The *agent* determines it needs more information and decides to use a *research* tool.
3. The *research* tool also contains a list of tools to facilitate planning. It contains a *web search* tool to gather information about travel to Japan, family-friendly activities, and typical costs.